DNN accelerator reliability
--Motivation
--Metrics
--Reliability characteristics

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Acronyms

- TE =timing error
- BER = bit error rate
- ECC = error-correcting code
- TPU = (Google's) tensor processing unit
- SECDED = single error correction, double error detection
- SDC = silent data corruption
- DUE = detected unrecoverable error
- TMR = triple modular redundancy
- FP = floating point, FxP = fixed point

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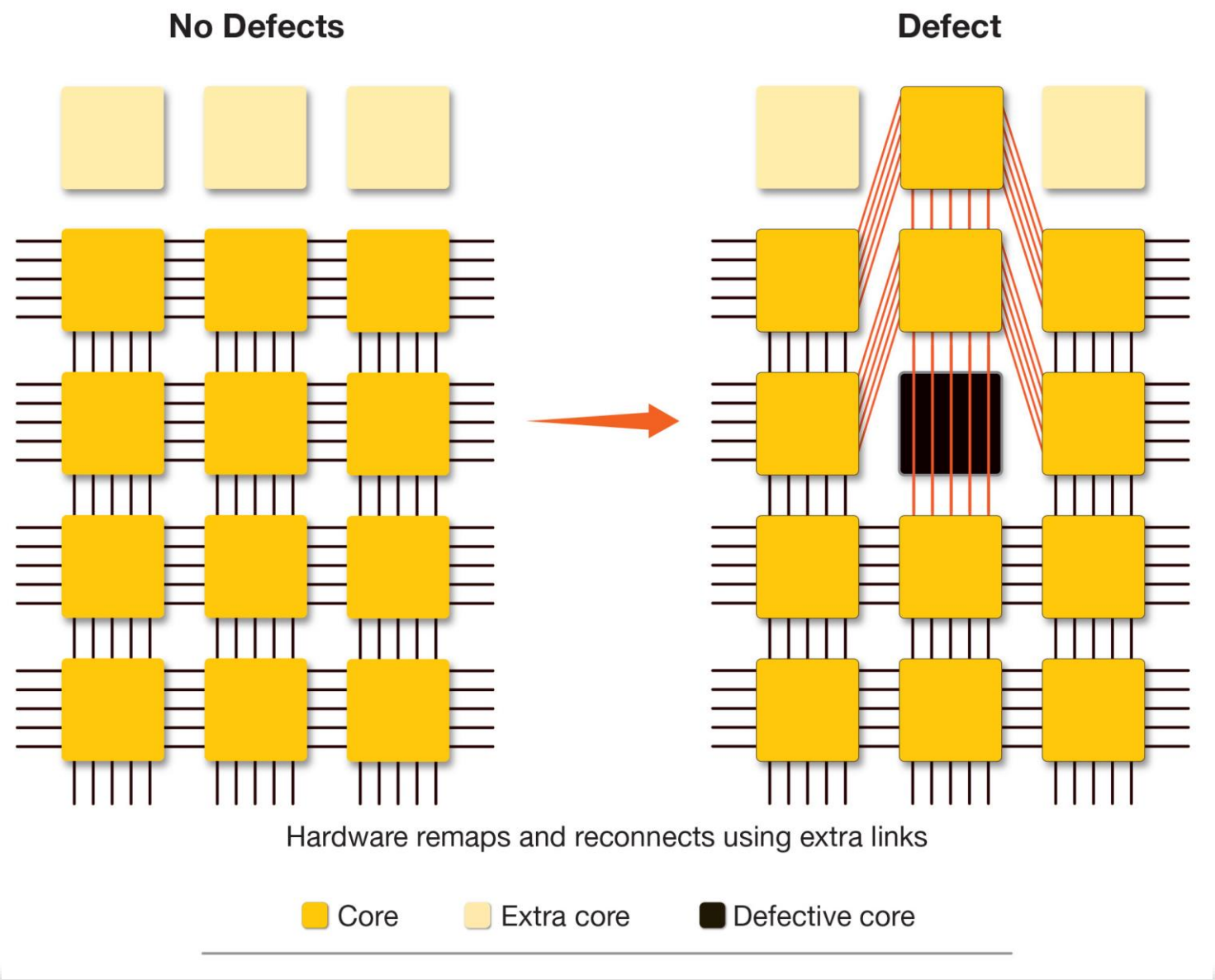
Motivation for improving reliability of DNN accelerators/algorithms

Importance of protecting DNN accelerators

- DNNs are being used for mission-critical applications
- Vulnerability values of different DNNs and even different layers of a single DNN vary widely
- Optimizing other metrics can compromise reliability
  - E.g., to achieve high data-reuse, the memory in DNN accelerators is accessed repeatedly, and due to this, a faulty value may be reused several times.
- Conventional reliability techniques, such as redundancy or re-computation do not take into account the characteristics of DNN accelerators and algorithms ➔ they may provide under-protection or over-protection

Example of redundancy in Cerebras WSE



Challenges of protecting DNN accelerator

- Generally, DNNs show a negligible loss in accuracy as long as the BER is below a certain threshold, but beyond this threshold, they show an exponential reduction in accuracy
- Multi-bit errors are on rise but most conventional techniques work for single-bit errors only
- Under aggressive optimizations such as voltage scaling and DRAM refresh rate scaling, the conventional reliability techniques may no longer be sufficient

Challenges of ECCs

- ECC not suitable for L1 cache
- Low-voltage => errors increase superlinearly => ECC becomes ineffective even for LLCs
- Necessitates stronger ECCs => high overhead

Limitations of ECC in DNNs

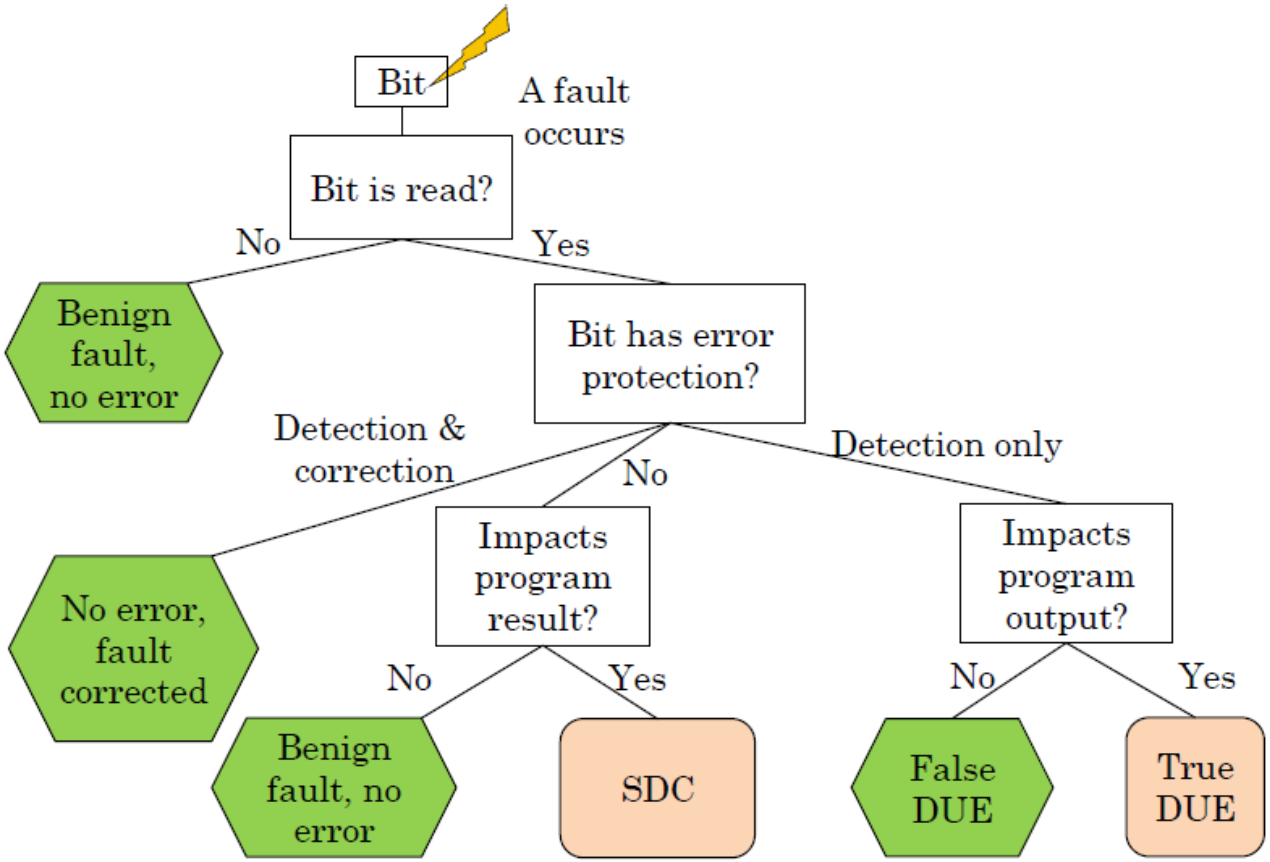
- ECC mitigates those errors that are mitigated by max-pooling layers also
- ECC cannot prevent rectangle and cubic errors which have the highest impact on classification accuracy
- SECDED ECC is ineffective in protecting against double-bit error
- In GPUs, ECC is used only for memory-elements, but the faults in remaining unprotected components have a higher probability of impacting final accuracy than the faults occurring in the ECC-protected memory components

Impact of increasing BER

- Generally, DNNs show a negligible loss in accuracy as long as the BER is below a certain threshold, but beyond this threshold, they show an exponential reduction in accuracy
- With increasing BER, not only more data-values of a DNN become faulty, but the magnitude of change in data-values also becomes high.
- Larger the magnitude of change in value due to a fault, higher is the probability of output error.

Metrics

Background: Fault taxonomy



Some metrics used (1 of 2)

- Classification accuracy (for image classification DNNs)
- Ability to detect the object (for object detection DNNs)
- Highest BER at which accuracy loss remains zero
- Percentage of parameters that are vulnerable (i.e., a bit-flip in it degrades accuracy severely)
- Percentage of trials that reach a certain accuracy. This is used when faults are introduced at random locations or in random weights and a large number of trials are performed to find statistically meaningful results
- Number of simultaneous bit flips required for making a DNN unstable
- Resilience of a neuron

Some metrics used (2 of 2)

- SDC rate and crash rate
- SDC-1 and SDC-5 where the highest-ranked item predicted by the DNN is not in the top one and top five (respectively) predictions of fault-free execution
- SDC-10% and SDC-20%, where confidence score of top-ranked element differs by more than 10% and 20% (respectively) of the fault-free execution.
- “Mean executions between failures'“: obtained by dividing the error rate by the execution time
- “Tolerable errors” are defined as those where the DNN output is faulty, but the classification result is unaffected and “critical errors”' as those where the classification result is also wrong

Reliability of different layers

- ReLU and max-pooling layers
- Normalization layers such as BN and LRN
- CONV and FC layers

Error-masking behavior of ReLU and max-pooling layers

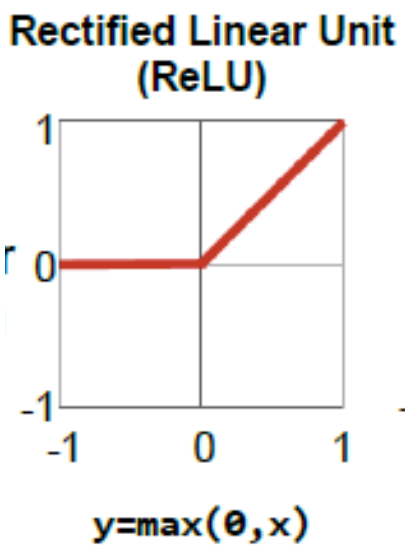
- Max-pooling: fault in non-maximum values => no impact

2	2	4	4
2	4	8	4
4	4	4	1
6	10	3	4

Max pooling

4	8
10	4

- fault in negative activations => no impact on using ReLU

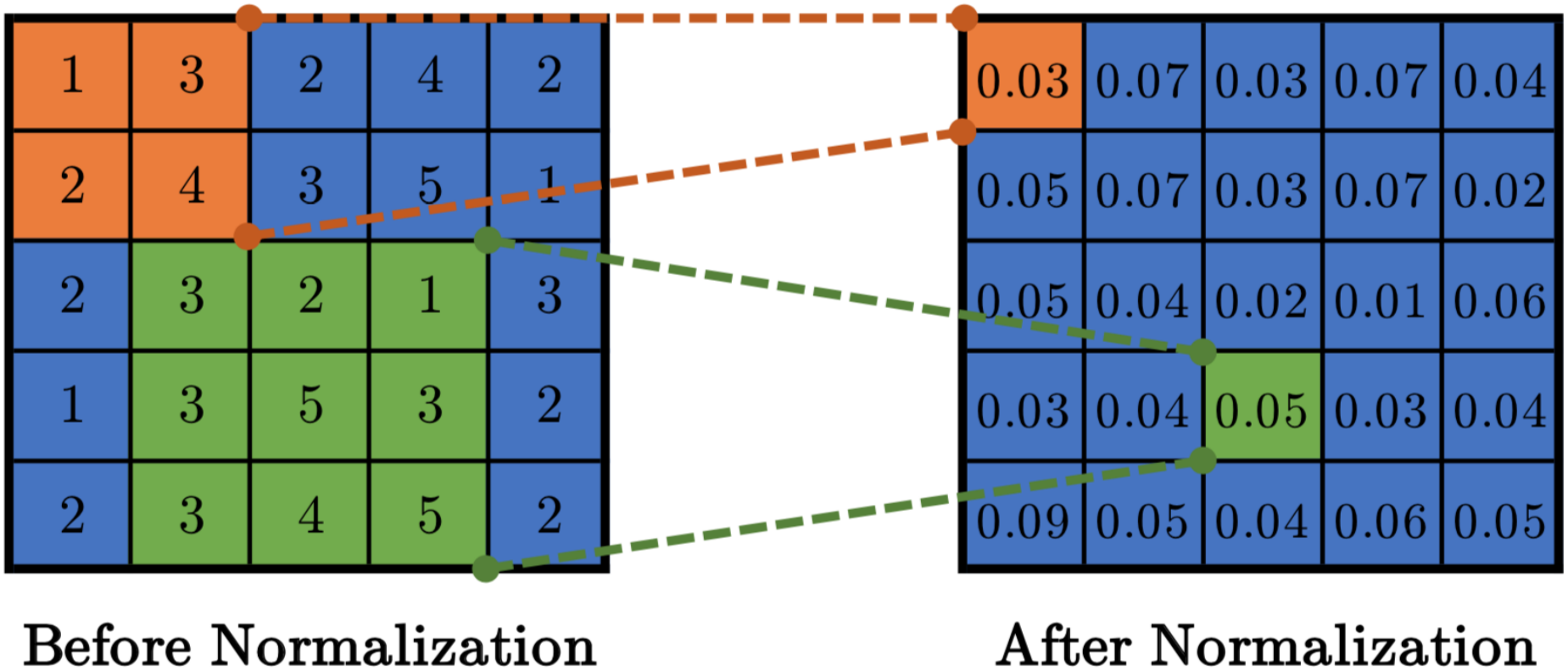


Error-masking behavior of normalization layers

- Normalization layers such as BN and LRN mask the faults
  - by normalizing the erroneous values and removing the impact of massive changes.
- Hence, other error-detectors can be placed after pooling/ReLU/normalization layers
- This avoids mitigating the faults that are suppressed by these layers.

BN = batch normalization
LRN = local response normalization

LRN



LRN square-normalizes the pixel values in a feature map within a local neighborhood.

Reliability of FC and CONV layers

- **FC:** Due to full-connection, a fault in an input activation to an FC layer spreads to all output activations.
- Since generally there is no ReLU or pooling layer after an FC layer, a fault in FC layer is not masked and affects the DNN output directly.
- **CONV:** Since the CONV layers are followed by pooling or ReLU layers, faults in CONV layers are masked.
- However, since the CONV layers are placed towards the beginning of a DNN, a fault in a CONV layer, if unmasked, can affect many layers of the DNN

Reliability of different data-structures and data formats

- Weights and activations
- FP vs fixed-point values

Reliability of weights/activations

- Activations are more robust than weights.
  - Possible optimization: Use ECC only for weights and not for activations

Reliability of FP vs FxP, and different FxP formats

- Compared to FxP, FP has larger dynamic range
- Hence, a fault in the exponent of FP can bring large change in its magnitude.
- Hence, for the same total bitwidth, FxP format is more robust than FP format.
- For the same reason, the robustness of a binary-quantized DNN is high

Reliability of different FxP formats

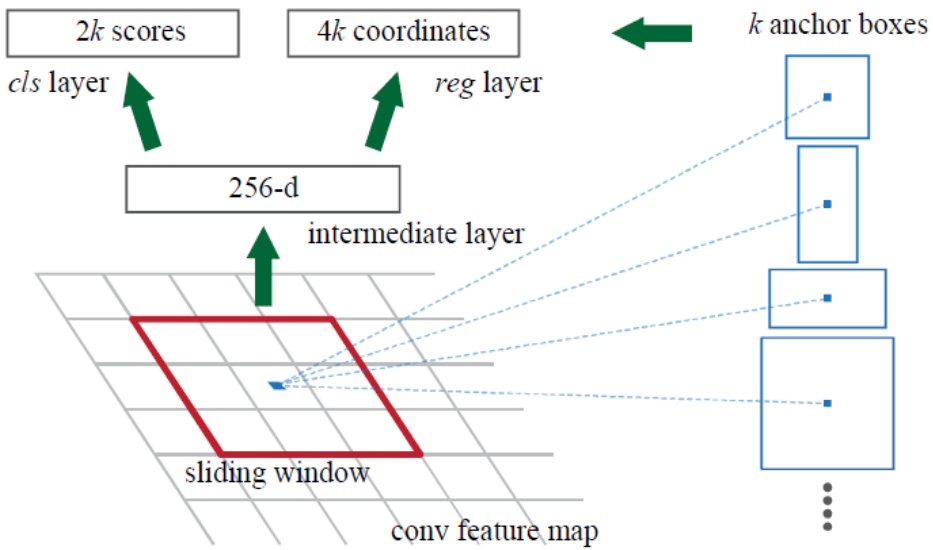
- For FxP formats with same total bitwidth, the one with less number of integer bits is more robust
- Reason: a fault in fraction bits has negligible impact but a fault in integer bits can bring a large change in the magnitude.

Impact of number of output labels on reliability

- **Note:** Since output of DNN is generally a ranking, an error which changes the confidence scores but does not change ranking of predicted class labels is not critical
- ImageNet has 1000 classes, CIFAR-10 has only 10
- For a dataset with **high** number of classes, an error in top-ranked elements can change the ranking to outside of the top-five class-labels.
- By comparison, in a dataset with **low** number of classes, even after occurrence of an error, the ranking may still remain within the top-five class-labels

Impact of output format on reliability (1 of 2)

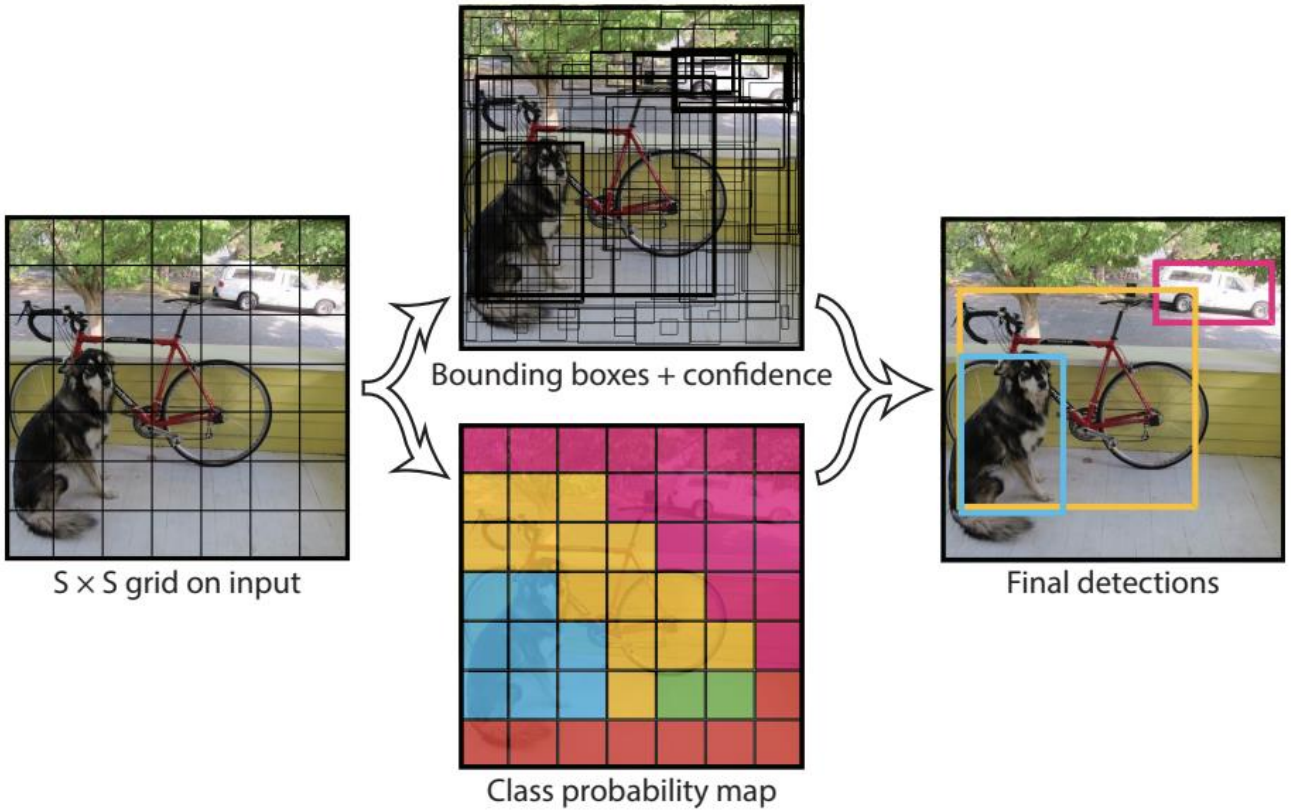
- Critical SDCs of Faster-RCNN << YOLO.
- Faster RCNN uses multiple anchor boxes for defining an object and thus, an error in one of the coordinates of a vertex does not alter the shape.



Region proposal network used in Faster-RCNN

Impact of output format on reliability (2 of 2)

- YOLO represents an object in terms of coordinates of its center, width, and height. Corruption of center-coordinates shifts the object location, and thus, the actual object is not detected



YOLO algorithm